

Hitec University Taxila, Department of Computer Engineering

Lab Manual for Mobile Application Development

Lab-8 Fragments

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1 Introduction

A Fragment is a piece of an activity which enable more modular activity design. A fragment is a kind of sub-activity. A Fragment represents a behavior or a portion of user interface in a Fragment Activity

2 Activity Time boxing

Table 1: Activity Time Boxing

Task#	Activity Name	Activity time	Total Time
1	Introduction	60mins	60min
2	Walk Through Task	60mins	60min
3	Evaluation Task	60min	60min

3 Objectives of the experiment

- To get basic understanding regarding fragments
- To get basic understanding regarding how fragments pass information between fragments
- To get to know about the benefits related to Fragments

4 Walk through Task

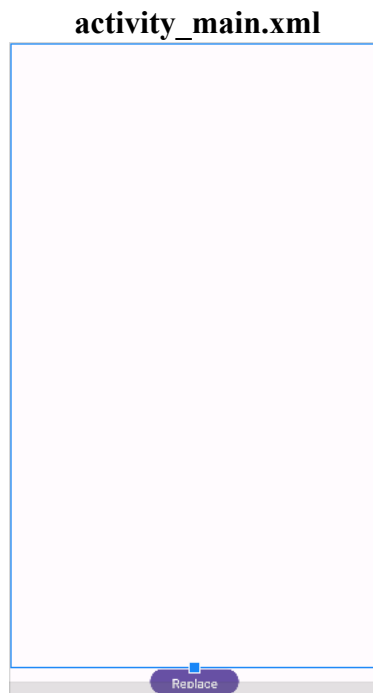
Creating New Activity

Step 1: Create Activities:

Create a new project just like you have created in your previous labs and create one more empty activity.

Step 2: Design XMLs of Activity:

Designs of activity are given below, just use the given xml code which is given in this step.



```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:weightSum="4"
    tools:context="Fragments.MainFragmentActivity">

    <LinearLayout
        android:orientation="vertical"
        android:id="@+id/frame_container"
        android:layout_weight="3"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
    </LinearLayout>

    <Button
        android:id="@+id/replace_btn"
        android:layout_weight="1"
        android:layout_gravity="center"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Replace" />
</LinearLayout>

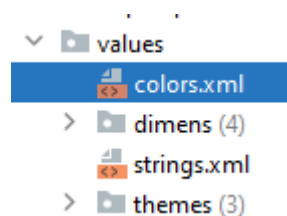
```

Step 3: Update colors.xml

```

<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="green">#FF00FF00</color>
    <color name="blue">#FF0000FF</color>
</resources>

```



Step 3: Write XML for two fragments



```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="@dimen/view_margin"
    android:background="@color/green"
    android:layout_width="match_parent"
    android:weightSum="10"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/fragment_title_a"
        android:gravity="center"
        android:textSize="32sp"
        android:textColor="@color/white"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="8"
        android:text="Fragment A" />

    <TextView
        android:id="@+id/main_message_a"
        android:gravity="center"
        android:textColor="@color/white"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="2"
        android:text="Message from main" />
</LinearLayout>
```



```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="@dimen/view_margin"
    android:background="@color/blue"
    android:layout_width="match_parent"
    android:weightSum="10"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/fragment_title_a"
        android:gravity="center"
        android:textSize="32sp"
        android:textColor="@color/white"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="8"
        android:text="Fragment B" />

    <TextView
        android:id="@+id/main_message_b_tv"
        android:gravity="center"
        android:textColor="@color/white"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="2"
        android:text="Message from main" />
</LinearLayout>
```

Step 4: Write fragments classes

FragmentA.java

```
public class FragmentA extends Fragment {
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable
        ViewGroup container, @Nullable Bundle savedInstanceState) {
        View fragmentView = inflater.inflate(R.layout.fragment_a_layout,
            container, false);
        TextView fragmentTV = fragmentView.findViewById(R.id.main_message_a_tv);
        if(getArguments() != null) {
            String msg= getArguments().getString(MainFragmentActivity.MSG_KEY);
            fragmentTV.setText(msg);
        }
        return fragmentView;
    }
}
```

FragmentB.java

```
public class FragmentB extends Fragment {
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable
        ViewGroup container, @Nullable Bundle savedInstanceState) {
        View fragmentView = inflater.inflate(R.layout.fragment_b_layout,
            container, false);
        TextView fragmentTV = fragmentView.findViewById(R.id.main_message_b_tv);
        if(getArguments() != null) {
            String msg= getArguments().getString(MainFragmentActivity.MSG_KEY);
            fragmentTV.setText(msg);
        }
        return fragmentView;
    }
}
```

Step 5: Write main activity class

```
public class MainFragmentActivity extends AppCompatActivity {

    boolean isFragmentA = true;
    FragmentManager fragmentManager;
    FragmentTransaction fragmentTransaction;
    FragmentA fragmentA;
    FragmentB fragmentB;
    String messageA = "Message from Main for Fragment A";
    String messageB = "Message from Main for Fragment B";
    Bundle args= new Bundle();
    public static final String MSG_KEY = "MSG_KEY";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main_fragment);

        fragmentA = new FragmentA();
        fragmentB = new FragmentB();

        replaceFragment(fragmentA, messageA);
    }
}
```

```

Button replaceBtn = findViewById(R.id.replace_btn);
replaceBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if(!isFragmentA) {
            replaceFragment(fragmentA, messageA);
            isFragmentA = true;
        } else {
            replaceFragment(fragmentB, messageB);
            isFragmentA = false;
        }
    }
});
}

void replaceFragment(Fragment fragment, String message){
    args.putString(MSG_KEY, message);
    fragment.setArguments(args);

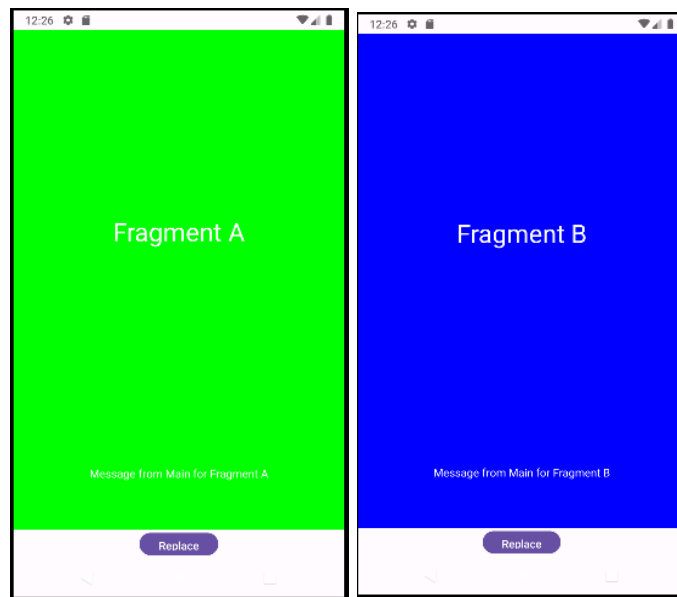
    fragmentManager = getSupportFragmentManager();
    fragmentTransaction = fragmentManager.beginTransaction();

    fragmentTransaction.replace(R.id.frame_container, fragment);
    fragmentTransaction.addToBackStack(null);
    fragmentTransaction.commit();
}
}

```

Step 6: Now Run Project

Click “Run” or press “Shift + F10”.



5 Home Task

Use Tasbeeh app and show history in separate fragment instead of separate activity.

Note: Bundle can share string array list